

Dashcam perceived risk (#239500)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 3 authors.

Pre-registered on: 2025/07/22 - 05:25 PM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

Does the type of driver (Autonomous Vehicle vs. Human-Driven Vehicle) affect passengers' perceived risk during vehicle-to-vehicle interaction moments? If so, when specifically?

3) Describe the key dependent variable(s) specifying how they will be measured.

Passenger's perceived risk in the driving situation, measured after each video clips using the item:
"As a passenger, how much risk did you perceive in the last scene?" (1 = Not at all, 7 = Extremely)

4) How many and which conditions will participants be assigned to?

We will use a one-factor, two-level between-subjects design:

1. Human-driven vehicle (HDV): Participants will be asked to imagine themselves as a passenger in a legally licensed taxi or ride-hailing vehicle.
2. Autonomous vehicle (AV): Participants will be asked to imagine themselves as a passenger in a legally licensed autonomous vehicle.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

The Causal Forest method will be used to estimate the Conditional Average Treatment Effect (CATE) of the autonomous vehicle (AV) driver setting. Following this, we will conduct a systematic subgroup analysis to identify key factors contributing to treatment effect heterogeneity. We will define subgroups using one-, two-, and three-way combinations of categorical features and compute the mean CATE for each group meeting a minimum size threshold. The output will present ranked lists of feature combinations associated with the largest positive and negative effects, thereby isolating the characteristics of the most and least responsive groups.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Participants will be excluded from the analysis if they fail either the manipulation check (i.e., their response does not match the assigned condition) or the attention check.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will recruit a total of 1,000 participants (500 per condition) from Prolific. Participants will be randomly assigned via Qualtrics to one of the 2 conditions.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Each participant will be randomly assigned to one of the two conditions and will view 10 randomly selected 5-second video clips (from a total pool of 1325 clips). After watching each video, participants will respond to the perceived risk question. Also, we will include control variables such as trust in human drivers (HDV trust) and autonomous vehicles (AV trust), prior experience with AVs, primary mode of transportation, frequency of ADAS use, technological confidence, driving experience, continent of residence, gender, and age.